Table 1: MNIST 6 vs 9 – Expanding Architecture ($k = 10$)

Layer	**Accuracy**	**Intra-class (κ_{intra})**	**Inter-class (κ_{inter})**	**Geometric Delta ($\Delta\kappa$)**
Layer 1	$83.33\% \pm 5.13\%$	0.3548 ± 0.0175	0.3206 ± 0.0206	0.0342 ± 0.0125
Layer 2	$83.78\% \pm 2.83\%$	0.3372 ± 0.0113	0.2945 ± 0.0097	0.0427 ± 0.0167
Layer 3	$97.33\% \pm 0.67\%$	0.3539 ± 0.0085	0.1871 ± 0.0147	0.1669 ± 0.0154
Layer 4	$99.89\% \pm 0.19\%$	0.4109 ± 0.0085	0.1006 ± 0.0609	0.3103 ± 0.0613
Layer 5	$99.89\% \pm 0.19\%$	0.4310 ± 0.0078	0.0760 ± 0.0289	0.3550 ± 0.0343

Table 2: MNIST 6 vs 9 – Expanding Architecture ($k = 15$)

Layer	**Accuracy**	**Intra-class (κ_{intra})**	**Inter-class (κ_{inter})**	**Geometric Delta ($\Delta\kappa$)**
Layer 1	$83.33\% \pm 5.13\%$	0.3535 ± 0.0143	0.3412 ± 0.0137	0.0123 ± 0.0021
Layer 2	$83.78\% \pm 2.83\%$	0.3433 ± 0.0114	0.3117 ± 0.0106	0.0316 ± 0.0138
Layer 3	$97.33\% \pm 0.67\%$	0.3503 ± 0.0089	0.1874 ± 0.0193	0.1629 ± 0.0211
Layer 4	$99.89\% \pm 0.19\%$	0.4137 ± 0.0064	0.0846 ± 0.0496	0.3291 ± 0.0543
Layer 5	$99.89\% \pm 0.19\%$	0.4313 ± 0.0049	0.1375 ± 0.0279	0.2938 ± 0.0237

Table 3: MNIST 6 vs 9 – Expanding Architecture ($k = 20$)

Layer	**Accuracy**	**Intra-class (κ_{intra})**	**Inter-class (κ_{inter})**	**Geometric Delta ($\Delta\kappa$)**
Layer 1	$83.33\% \pm 5.13\%$	0.3619 ± 0.0111	0.3607 ± 0.0133	0.0013 ± 0.0041
Layer 2	$83.78\% \pm 2.83\%$	0.3569 ± 0.0070	0.3280 ± 0.0108	0.0289 ± 0.0112
Layer 3	$97.33\% \pm 0.67\%$	0.3560 ± 0.0060	0.1985 ± 0.0174	0.1575 ± 0.0202
Layer 4	$99.89\% \pm 0.19\%$	0.4295 ± 0.0043	0.1006 ± 0.0510	0.3289 ± 0.0525
Layer 5	$99.89\% \pm 0.19\%$	0.4440 ± 0.0032	0.0965 ± 0.0080	0.3475 ± 0.0066

Table 4: MNIST 6 vs 9 – Bottleneck Architecture ($k = 10$)

Layer	**Accuracy**	**Intra-class (κ_{intra})**	**Inter-class (κ_{inter})**	**Geometric Delta ($\Delta\kappa$)**
Layer 1	$79.11\% \pm 6.00\%$	0.3470 ± 0.0038	0.3412 ± 0.0149	0.0058 ± 0.0123
Layer 2	$84.22\% \pm 3.40\%$	0.3445 ± 0.0117	0.2976 ± 0.0138	0.0468 ± 0.0022
Layer 3	$98.11\% \pm 1.07\%$	0.3829 ± 0.0033	0.2360 ± 0.0108	0.1469 ± 0.0080
Layer 4	$99.22\% \pm 0.51\%$	0.3974 ± 0.0042	0.2365 ± 0.0685	0.1609 ± 0.0717
Layer 5	$99.78\% \pm 0.38\%$	0.4015 ± 0.0086	0.2008 ± 0.0443	0.2006 ± 0.0526

Table 5: MNIST 6 vs 9 – Bottleneck Architecture ($k = 15$)

Layer	**Accuracy**	**Intra-class (κ_{intra})**	**Inter-class (κ_{inter})**	**Geometric Delta ($\Delta\kappa$)**
Layer 1	$79.11\% \pm 6.00\%$	0.3540 ± 0.0097	0.3491 ± 0.0155	0.0049 ± 0.0072
Layer 2	$84.22\% \pm 3.40\%$	0.3520 ± 0.0089	0.3120 ± 0.0102	0.0400 ± 0.0014
Layer 3	$98.11\% \pm 1.07\%$	0.3881 ± 0.0073	0.2286 ± 0.0040	0.1595 ± 0.0089
Layer 4	$99.22\% \pm 0.51\%$	0.4059 ± 0.0108	0.2278 ± 0.0538	0.1781 ± 0.0644
Layer 5	$99.78\% \pm 0.38\%$	0.4134 ± 0.0127	0.1792 ± 0.0513	0.2342 ± 0.0593

Table 6: MNIST 6 vs 9 – Bottleneck Architecture ($k = 20$)

Layer	**Accuracy**	**Intra-class (κ_{intra})**	**Inter-class (κ_{inter})**	**Geometric Delta ($\Delta\kappa$)**
Layer 1	$79.11\% \pm 6.00\%$	0.3640 ± 0.0127	0.3655 ± 0.0131	-0.0015 ± 0.0004
Layer 2	$84.22\% \pm 3.40\%$	0.3642 ± 0.0044	0.3309 ± 0.0110	0.0333 ± 0.0066
Layer 3	$98.11\% \pm 1.07\%$	0.3983 ± 0.0061	0.2462 ± 0.0023	0.1521 ± 0.0050
Layer 4	$99.22\% \pm 0.51\%$	0.4212 ± 0.0127	0.2259 ± 0.0240	0.1953 ± 0.0367
Layer 5	$99.78\% \pm 0.38\%$	0.4296 ± 0.0126	0.2084 ± 0.0327	0.2212 ± 0.0379

Table 7: MNIST 6 vs 9 – Collapsed Tower Architecture ($k = 10$)

Layer	**Accuracy**	**Intra-class (κ_{intra})**	**Inter-class (κ_{inter})**	**Geometric Delta ($\Delta\kappa$)**
Layer 1	$84.11\% \pm 3.47\%$	0.3370 ± 0.0055	0.3126 ± 0.0083	0.0244 ± 0.0113
Layer 2	$84.67\% \pm 3.33\%$	0.3231 ± 0.0019	0.2916 ± 0.0104	0.0315 ± 0.0084
Layer 3	$96.89\% \pm 0.51\%$	0.3692 ± 0.0046	0.2300 ± 0.0052	0.1392 ± 0.0098
Layer 4	$99.00\% \pm 0.33\%$	0.3778 ± 0.0083	0.1950 ± 0.0126	0.1828 ± 0.0180
Layer 5	$99.33\% \pm 0.33\%$	0.3888 ± 0.0130	0.1934 ± 0.0312	0.1954 ± 0.0443

Table 8: MNIST 6 vs 9 – Collapsed Tower Architecture ($k = 15$)

Layer	**Accuracy**	**Intra-class (κ_{intra})**	**Inter-class (κ_{inter})**	**Geometric Delta ($\Delta\kappa$)**
Layer 1	$84.11\% \pm 3.47\%$	0.3440 ± 0.0063	0.3351 ± 0.0094	0.0089 ± 0.0120
Layer 2	$84.67\% \pm 3.33\%$	0.3353 ± 0.0047	0.3022 ± 0.0142	0.0331 ± 0.0126
Layer 3	$96.89\% \pm 0.51\%$	0.3740 ± 0.0117	0.2320 ± 0.0043	0.1421 ± 0.0086
Layer 4	$99.00\% \pm 0.33\%$	0.3848 ± 0.0081	0.1776 ± 0.0159	0.2072 ± 0.0151
Layer 5	$99.33\% \pm 0.33\%$	0.4032 ± 0.0135	0.1840 ± 0.0479	0.2192 ± 0.0613

Table 9: MNIST 6 vs 9 – Collapsed Tower Architecture ($k = 20$)

Layer	**Accuracy**	**Intra-class (κ_{intra})**	**Inter-class (κ_{inter})**	**Geometric Delta ($\Delta\kappa$)**
Layer 1	$84.11\% \pm 3.47\%$	0.3563 ± 0.0063	0.3503 ± 0.0088	0.0061 ± 0.0102
Layer 2	$84.67\% \pm 3.33\%$	0.3464 ± 0.0052	0.3192 ± 0.0156	0.0272 ± 0.0105
Layer 3	$96.89\% \pm 0.51\%$	0.3867 ± 0.0162	0.2452 ± 0.0104	0.1415 ± 0.0097
Layer 4	$99.00\% \pm 0.33\%$	0.4022 ± 0.0062	0.1981 ± 0.0134	0.2040 ± 0.0100
Layer 5	$99.33\% \pm 0.33\%$	0.4266 ± 0.0196	0.1925 ± 0.0613	0.2340 ± 0.0802

Table 10: MNIST 1 vs 7 – Expanding Architecture ($k = 10$)

Layer	**Accuracy**	**Intra-class (κ_{intra})**	**Inter-class (κ_{inter})**	**Geometric Delta ($\Delta\kappa$)**
Layer 1	$98.67\% \pm 0.88\%$	0.4076 ± 0.0115	0.3210 ± 0.0464	0.0865 ± 0.0551
Layer 2	$98.22\% \pm 0.51\%$	0.4103 ± 0.0121	0.2685 ± 0.0492	0.1418 ± 0.0495
Layer 3	$97.00\% \pm 0.58\%$	0.4418 ± 0.0160	0.3115 ± 0.0144	0.1303 ± 0.0260
Layer 4	$99.67\% \pm 0.33\%$	0.4692 ± 0.0197	0.2537 ± 0.0710	0.2155 ± 0.0812
Layer 5	$99.78\% \pm 0.19\%$	0.4634 ± 0.0145	0.1908 ± 0.1218	0.2726 ± 0.1325

Table 11: MNIST 1 vs 7 – Expanding Architecture ($k = 15$)

Layer	**Accuracy**	**Intra-class (κ_{intra})**	**Inter-class (κ_{inter})**	**Geometric Delta ($\Delta\kappa$)**
Layer 1	$98.67\% \pm 0.88\%$	0.4254 ± 0.0086	0.3060 ± 0.0174	0.1194 ± 0.0152
Layer 2	$98.22\% \pm 0.51\%$	0.4273 ± 0.0097	0.2524 ± 0.0478	0.1749 ± 0.0481
Layer 3	$97.00\% \pm 0.58\%$	0.4428 ± 0.0125	0.2994 ± 0.0160	0.1434 ± 0.0261
Layer 4	$99.67\% \pm 0.33\%$	0.4769 ± 0.0158	0.2465 ± 0.0540	0.2304 ± 0.0576
Layer 5	$99.78\% \pm 0.19\%$	0.4760 ± 0.0142	0.1932 ± 0.1014	0.2828 ± 0.1131

Table 12: MNIST 1 vs 7 – Expanding Architecture ($k = 20$)

Layer	**Accuracy**	**Intra-class (κ_{intra})**	**Inter-class (κ_{inter})**	**Geometric Delta ($\Delta\kappa$)**
Layer 1	$98.67\% \pm 0.88\%$	0.4418 ± 0.0072	0.3016 ± 0.0373	0.1403 ± 0.0376
Layer 2	$98.22\% \pm 0.51\%$	0.4445 ± 0.0083	0.2475 ± 0.0495	0.1970 ± 0.0500
Layer 3	$97.00\% \pm 0.58\%$	0.4471 ± 0.0056	0.3073 ± 0.0112	0.1398 ± 0.0167
Layer 4	$99.67\% \pm 0.33\%$	0.4894 ± 0.0126	0.2541 ± 0.0506	0.2353 ± 0.0510
Layer 5	$99.78\% \pm 0.19\%$	0.4916 ± 0.0117	0.2329 ± 0.0827	0.2587 ± 0.0920

Table 13: MNIST 1 vs 7 – Bottleneck Architecture ($k = 10$)

Layer	**Accuracy**	**Intra-class (κ_{intra})**	**Inter-class (κ_{inter})**	**Geometric Delta ($\Delta\kappa$)**
Layer 1	$97.78\% \pm 0.51\%$	0.4143 ± 0.0117	0.3208 ± 0.0180	0.0935 ± 0.0082
Layer 2	$96.89\% \pm 0.84\%$	0.4233 ± 0.0087	0.3067 ± 0.0208	0.1166 ± 0.0280
Layer 3	$99.33\% \pm 0.33\%$	0.4325 ± 0.0076	0.2758 ± 0.0469	0.1567 ± 0.0545
Layer 4	$99.22\% \pm 0.38\%$	0.4218 ± 0.0071	0.2609 ± 0.0368	0.1609 ± 0.0302
Layer 5	$99.44\% \pm 0.51\%$	0.4319 ± 0.0130	0.3004 ± 0.0193	0.1315 ± 0.0072

Table 14: MNIST 1 vs 7 – Bottleneck Architecture ($k = 15$)

Layer	**Accuracy**	**Intra-class (κ_{intra})**	**Inter-class (κ_{inter})**	**Geometric Delta ($\Delta\kappa$)**
Layer 1	$97.78\% \pm 0.51\%$	0.4202 ± 0.0044	0.3076 ± 0.0458	0.1126 ± 0.0431
Layer 2	$96.89\% \pm 0.84\%$	0.4311 ± 0.0051	0.3028 ± 0.0142	0.1283 ± 0.0172
Layer 3	$99.33\% \pm 0.33\%$	0.4362 ± 0.0009	0.2826 ± 0.0218	0.1536 ± 0.0226
Layer 4	$99.22\% \pm 0.38\%$	0.4315 ± 0.0053	0.2835 ± 0.0155	0.1481 ± 0.0166
Layer 5	$99.44\% \pm 0.51\%$	0.4369 ± 0.0098	0.3483 ± 0.0346	0.0886 ± 0.0267

Table 15: MNIST 1 vs 7 – Bottleneck Architecture ($k = 20$)

Layer	**Accuracy**	**Intra-class (κ_{intra})**	**Inter-class (κ_{inter})**	**Geometric Delta ($\Delta\kappa$)**
Layer 1	$97.78\% \pm 0.51\%$	0.4350 ± 0.0077	0.3080 ± 0.0272	0.1271 ± 0.0274
Layer 2	$96.89\% \pm 0.84\%$	0.4463 ± 0.0051	0.3088 ± 0.0094	0.1375 ± 0.0142
Layer 3	$99.33\% \pm 0.33\%$	0.4488 ± 0.0063	0.2974 ± 0.0224	0.1515 ± 0.0286
Layer 4	$99.22\% \pm 0.38\%$	0.4467 ± 0.0028	0.3062 ± 0.0201	0.1405 ± 0.0203
Layer 5	$99.44\% \pm 0.51\%$	0.4521 ± 0.0079	0.3766 ± 0.0128	0.0755 ± 0.0061

Table 16: MNIST 1 vs 7 – Collapsed Architecture ($k = 10$)

Layer	**Accuracy**	**Intra-class (κ_{intra})**	**Inter-class (κ_{inter})**	**Geometric Delta ($\Delta\kappa$)**
Layer 1	$97.33\% \pm 1.00\%$	0.3984 ± 0.0097	0.2802 ± 0.0050	0.1181 ± 0.0059
Layer 2	$97.56\% \pm 0.19\%$	0.4028 ± 0.0056	0.3032 ± 0.0179	0.0996 ± 0.0235
Layer 3	$98.78\% \pm 0.38\%$	0.4142 ± 0.0042	0.2335 ± 0.0346	0.1807 ± 0.0304
Layer 4	$98.67\% \pm 0.58\%$	0.4172 ± 0.0057	0.2500 ± 0.0240	0.1672 ± 0.0184
Layer 5	$99.11\% \pm 0.19\%$	0.4282 ± 0.0060	0.3507 ± 0.0401	0.0774 ± 0.0454

Table 17: MNIST 1 vs 7 – Collapsed Architecture ($k = 15$)

Layer	**Accuracy**	**Intra-class (κ_{intra})**	**Inter-class (κ_{inter})**	**Geometric Delta ($\Delta\kappa$)**
Layer 1	$97.33\% \pm 1.00\%$	0.4105 ± 0.0096	0.2744 ± 0.0073	0.1360 ± 0.0037
Layer 2	$97.56\% \pm 0.19\%$	0.4117 ± 0.0012	0.2766 ± 0.0120	0.1351 ± 0.0130
Layer 3	$98.78\% \pm 0.38\%$	0.4166 ± 0.0033	0.2397 ± 0.0306	0.1770 ± 0.0301
Layer 4	$98.67\% \pm 0.58\%$	0.4195 ± 0.0062	0.2716 ± 0.0228	0.1479 ± 0.0167
Layer 5	$99.11\% \pm 0.19\%$	0.4299 ± 0.0059	0.2878 ± 0.0691	0.1421 ± 0.0653

Table 18: MNIST 1 vs 7 – Collapsed Architecture ($k = 20$)

Layer	**Accuracy**	**Intra-class (κ_{intra})**	**Inter-class (κ_{inter})**	**Geometric Delta ($\Delta\kappa$)**
Layer 1	$97.33\% \pm 1.00\%$	0.4276 ± 0.0106	0.2787 ± 0.0278	0.1489 ± 0.0184
Layer 2	$97.56\% \pm 0.19\%$	0.4273 ± 0.0046	0.2738 ± 0.0233	0.1535 ± 0.0279
Layer 3	$98.78\% \pm 0.38\%$	0.4323 ± 0.0054	0.2460 ± 0.0402	0.1863 ± 0.0418
Layer 4	$98.67\% \pm 0.58\%$	0.4357 ± 0.0068	0.2875 ± 0.0298	0.1482 ± 0.0235
Layer 5	$99.11\% \pm 0.19\%$	0.4421 ± 0.0051	0.3049 ± 0.0535	0.1372 ± 0.0506

Table 19: FASHION-MNIST Bags vs Dresses – Expanding Architecture ($k = 10$)

Layer	**Accuracy**	**Intra-class (κ_{intra})**	**Inter-class (κ_{inter})**	**Geometric Delta ($\Delta\kappa$)**
Layer 1	$94.56\% \pm 0.51\%$	0.4004 ± 0.0067	0.2812 ± 0.0131	0.1192 ± 0.0154
Layer 2	$96.00\% \pm 2.19\%$	0.4055 ± 0.0028	0.2478 ± 0.0141	0.1577 ± 0.0168
Layer 3	$96.78\% \pm 0.38\%$	0.4404 ± 0.0142	0.2964 ± 0.0231	0.1440 ± 0.0314
Layer 4	$98.89\% \pm 0.69\%$	0.4864 ± 0.0112	0.3039 ± 0.0722	0.1825 ± 0.0827
Layer 5	$99.33\% \pm 0.67\%$	0.4884 ± 0.0055	0.1827 ± 0.0162	0.3056 ± 0.0178

Table 20: FASHION-MNIST Bags vs Dresses – Expanding Architecture ($k = 15$)

Layer	**Accuracy**	**Intra-class (κ_{intra})**	**Inter-class (κ_{inter})**	**Geometric Delta ($\Delta\kappa$)**
Layer 1	$94.56\% \pm 0.51\%$	0.4110 ± 0.0043	0.2854 ± 0.0334	0.1255 ± 0.0370
Layer 2	$96.00\% \pm 2.19\%$	0.4202 ± 0.0057	0.2518 ± 0.0213	0.1684 ± 0.0264
Layer 3	$96.78\% \pm 0.38\%$	0.4503 ± 0.0102	0.3061 ± 0.0146	0.1442 ± 0.0180
Layer 4	$98.89\% \pm 0.69\%$	0.4945 ± 0.0098	0.2895 ± 0.0218	0.2050 ± 0.0312
Layer 5	$99.33\% \pm 0.67\%$	0.4995 ± 0.0047	0.1847 ± 0.0165	0.3148 ± 0.0212

Table 21: FASHION-MNIST Bags vs Dresses – Expanding Architecture ($k = 20$)

Layer	**Accuracy**	**Intra-class (κ_{intra})**	**Inter-class (κ_{inter})**	**Geometric Delta ($\Delta\kappa$)**
Layer 1	$94.56\% \pm 0.51\%$	0.4281 ± 0.0028	0.2859 ± 0.0343	0.1421 ± 0.0350
Layer 2	$96.00\% \pm 2.19\%$	0.4362 ± 0.0104	0.2578 ± 0.0210	0.1784 ± 0.0308
Layer 3	$96.78\% \pm 0.38\%$	0.4665 ± 0.0106	0.3114 ± 0.0094	0.1552 ± 0.0137
Layer 4	$98.89\% \pm 0.69\%$	0.5133 ± 0.0131	0.2775 ± 0.0281	0.2358 ± 0.0389
Layer 5	$99.33\% \pm 0.67\%$	0.5178 ± 0.0065	0.1775 ± 0.0158	0.3402 ± 0.0218

Table 22: FASHION-MNIST Bags vs Dresses – Bottleneck Architecture ($k = 10$)

Layer	**Accuracy**	**Intra-class (κ_{intra})**	**Inter-class (κ_{inter})**	**Geometric Delta ($\Delta\kappa$)**
Layer 1	$94.33\% \pm 1.00\%$	0.3883 ± 0.0068	0.2966 ± 0.0334	0.0917 ± 0.0317
Layer 2	$95.78\% \pm 1.35\%$	0.4042 ± 0.0009	0.2873 ± 0.0388	0.1169 ± 0.0396
Layer 3	$98.00\% \pm 0.88\%$	0.4313 ± 0.0050	0.2800 ± 0.0362	0.1513 ± 0.0343
Layer 4	$98.56\% \pm 0.38\%$	0.4309 ± 0.0132	0.2787 ± 0.0651	0.1522 ± 0.0586
Layer 5	$98.33\% \pm 0.88\%$	0.4376 ± 0.0062	0.2979 ± 0.0512	0.1398 ± 0.0452

Table 23: FASHION-MNIST Bags vs Dresses – Bottleneck Architecture ($k = 15$)

Layer	**Accuracy**	**Intra-class (κ_{intra})**	**Inter-class (κ_{inter})**	**Geometric Delta ($\Delta\kappa$)**
Layer 1	$94.33\% \pm 1.00\%$	0.4016 ± 0.0092	0.3082 ± 0.0197	0.0934 ± 0.0126
Layer 2	$95.78\% \pm 1.35\%$	0.4183 ± 0.0100	0.2712 ± 0.0271	0.1472 ± 0.0188
Layer 3	$98.00\% \pm 0.88\%$	0.4505 ± 0.0053	0.2719 ± 0.0249	0.1787 ± 0.0236
Layer 4	$98.56\% \pm 0.38\%$	0.4532 ± 0.0092	0.2791 ± 0.0611	0.1741 ± 0.0519
Layer 5	$98.33\% \pm 0.88\%$	0.4573 ± 0.0159	0.2891 ± 0.0351	0.1681 ± 0.0326

Table 24: FASHION-MNIST Bags vs Dresses – Bottleneck Architecture ($k = 20$)

Layer	**Accuracy**	**Intra-class (κ_{intra})**	**Inter-class (κ_{inter})**	**Geometric Delta ($\Delta\kappa$)**
Layer 1	$94.33\% \pm 1.00\%$	0.4218 ± 0.0098	0.3217 ± 0.0164	0.1001 ± 0.0076
Layer 2	$95.78\% \pm 1.35\%$	0.4304 ± 0.0096	0.2803 ± 0.0198	0.1501 ± 0.0114
Layer 3	$98.00\% \pm 0.88\%$	0.4767 ± 0.0094	0.2875 ± 0.0121	0.1892 ± 0.0128
Layer 4	$98.56\% \pm 0.38\%$	0.4759 ± 0.0126	0.2843 ± 0.0324	0.1916 ± 0.0203
Layer 5	$98.33\% \pm 0.88\%$	0.4787 ± 0.0196	0.3017 ± 0.0181	0.1769 ± 0.0168

Table 25: FASHION-MNIST Bags vs Dresses – Collapsed Architecture ($k = 10$)

Layer	**Accuracy**	**Intra-class (κ_{intra})**	**Inter-class (κ_{inter})**	**Geometric Delta ($\Delta\kappa$)**
Layer 1	$94.33\% \pm 1.86\%$	0.3969 ± 0.0059	0.2896 ± 0.0120	0.1073 ± 0.0102
Layer 2	$95.11\% \pm 1.50\%$	0.4050 ± 0.0114	0.2571 ± 0.0288	0.1479 ± 0.0183
Layer 3	$97.78\% \pm 1.07\%$	0.4328 ± 0.0088	0.2971 ± 0.0208	0.1357 ± 0.0120
Layer 4	$98.67\% \pm 1.00\%$	0.4440 ± 0.0064	0.3411 ± 0.0157	0.1029 ± 0.0107
Layer 5	$98.56\% \pm 0.38\%$	0.4442 ± 0.0136	0.2829 ± 0.0272	0.1614 ± 0.0323

Table 26: FASHION-MNIST Bags vs Dresses – Collapsed Architecture ($k = 15$)

Layer	**Accuracy**	**Intra-class (κ_{intra})**	**Inter-class (κ_{inter})**	**Geometric Delta ($\Delta\kappa$)**
Layer 1	$94.33\% \pm 1.86\%$	0.4064 ± 0.0035	0.2937 ± 0.0150	0.1126 ± 0.0151
Layer 2	$95.11\% \pm 1.50\%$	0.4188 ± 0.0102	0.2572 ± 0.0241	0.1616 ± 0.0145
Layer 3	$97.78\% \pm 1.07\%$	0.4468 ± 0.0144	0.2768 ± 0.0263	0.1699 ± 0.0182
Layer 4	$98.67\% \pm 1.00\%$	0.4589 ± 0.0083	0.3113 ± 0.0310	0.1476 ± 0.0245
Layer 5	$98.56\% \pm 0.38\%$	0.4590 ± 0.0122	0.2778 ± 0.0132	0.1812 ± 0.0228

Table 27: FASHION-MNIST Bags vs Dresses – Collapsed Architecture ($k = 20$)

Layer	**Accuracy**	**Intra-class (κ_{intra})**	**Inter-class (κ_{inter})**	**Geometric Delta ($\Delta\kappa$)**
Layer 1	$94.33\% \pm 1.86\%$	0.4219 ± 0.0064	0.3071 ± 0.0193	0.1148 ± 0.0256
Layer 2	$95.11\% \pm 1.50\%$	0.4380 ± 0.0097	0.2698 ± 0.0169	0.1682 ± 0.0152
Layer 3	$97.78\% \pm 1.07\%$	0.4711 ± 0.0165	0.2932 ± 0.0225	0.1779 ± 0.0198
Layer 4	$98.67\% \pm 1.00\%$	0.4788 ± 0.0103	0.2845 ± 0.0158	0.1944 ± 0.0101
Layer 5	$98.56\% \pm 0.38\%$	0.4814 ± 0.0140	0.2850 ± 0.0231	0.1964 ± 0.0183

Table 28: FASHION-MNIST Sandals vs Sneakers – Expanding Architecture ($k = 10$)

Layer	**Accuracy**	**Intra-class (κ_{intra})**	**Inter-class (κ_{inter})**	**Geometric Delta ($\Delta\kappa$)**
Layer 1	$89.22\% \pm 2.83\%$	0.3893 ± 0.0092	0.3219 ± 0.0022	0.0674 ± 0.0091
Layer 2	$90.11\% \pm 1.35\%$	0.3974 ± 0.0113	0.3254 ± 0.0161	0.0720 ± 0.0238
Layer 3	$88.00\% \pm 0.58\%$	0.4272 ± 0.0190	0.2997 ± 0.0113	0.1275 ± 0.0267
Layer 4	$89.22\% \pm 1.17\%$	0.4469 ± 0.0199	0.3477 ± 0.0333	0.0992 ± 0.0138
Layer 5	$87.67\% \pm 0.00\%$	0.4557 ± 0.0269	0.3045 ± 0.0429	0.1512 ± 0.0197

Table 29: FASHION-MNIST Sandals vs Sneakers – Expanding Architecture ($k = 15$)

Layer	**Accuracy**	**Intra-class (κ_{intra})**	**Inter-class (κ_{inter})**	**Geometric Delta ($\Delta\kappa$)**
Layer 1	$89.22\% \pm 2.83\%$	0.4041 ± 0.0059	0.3269 ± 0.0045	0.0772 ± 0.0048
Layer 2	$90.11\% \pm 1.35\%$	0.4086 ± 0.0118	0.3277 ± 0.0136	0.0810 ± 0.0143
Layer 3	$88.00\% \pm 0.58\%$	0.4328 ± 0.0176	0.3109 ± 0.0230	0.1219 ± 0.0328
Layer 4	$89.22\% \pm 1.17\%$	0.4584 ± 0.0205	0.3421 ± 0.0300	0.1162 ± 0.0117
Layer 5	$87.67\% \pm 0.00\%$	0.4651 ± 0.0308	0.2971 ± 0.0348	0.1680 ± 0.0093

Table 30: FASHION-MNIST Sandals vs Sneakers – Expanding Architecture ($k = 20$)

Layer	**Accuracy**	**Intra-class (κ_{intra})**	**Inter-class (κ_{inter})**	**Geometric Delta ($\Delta\kappa$)**
Layer 1	$89.22\% \pm 2.83\%$	0.4164 ± 0.0107	0.3420 ± 0.0116	0.0744 ± 0.0104
Layer 2	$90.11\% \pm 1.35\%$	0.4265 ± 0.0103	0.3417 ± 0.0194	0.0848 ± 0.0157
Layer 3	$88.00\% \pm 0.58\%$	0.4421 ± 0.0107	0.3296 ± 0.0167	0.1125 ± 0.0215
Layer 4	$89.22\% \pm 1.17\%$	0.4743 ± 0.0149	0.3536 ± 0.0208	0.1207 ± 0.0068
Layer 5	$87.67\% \pm 0.00\%$	0.4811 ± 0.0230	0.3015 ± 0.0328	0.1795 ± 0.0104

Table 31: FASHION-MNIST Sandals vs Sneakers – Bottleneck Architecture ($k = 10$)

Layer	**Accuracy**	**Intra-class (κ_{intra})**	**Inter-class (κ_{inter})**	**Geometric Delta ($\Delta\kappa$)**
Layer 1	$88.11\% \pm 2.14\%$	0.4053 ± 0.0070	0.3308 ± 0.0213	0.0745 ± 0.0159
Layer 2	$90.67\% \pm 1.45\%$	0.4074 ± 0.0056	0.3298 ± 0.0122	0.0776 ± 0.0177
Layer 3	$93.89\% \pm 2.59\%$	0.4328 ± 0.0098	0.3243 ± 0.0266	0.1085 ± 0.0241
Layer 4	$94.89\% \pm 2.04\%$	0.4309 ± 0.0099	0.3138 ± 0.0319	0.1171 ± 0.0269
Layer 5	$95.33\% \pm 1.15\%$	0.4296 ± 0.0048	0.3098 ± 0.0333	0.1198 ± 0.0292

Table 32: FASHION-MNIST Sandals vs Sneakers – Bottleneck Architecture ($k = 15$)

Layer	**Accuracy**	**Intra-class (κ_{intra})**	**Inter-class (κ_{inter})**	**Geometric Delta ($\Delta\kappa$)**
Layer 1	$88.11\% \pm 2.14\%$	0.4160 ± 0.0148	0.3380 ± 0.0109	0.0780 ± 0.0039
Layer 2	$90.67\% \pm 1.45\%$	0.4219 ± 0.0069	0.3391 ± 0.0085	0.0827 ± 0.0125
Layer 3	$93.89\% \pm 2.59\%$	0.4386 ± 0.0036	0.3340 ± 0.0162	0.1046 ± 0.0164
Layer 4	$94.89\% \pm 2.04\%$	0.4377 ± 0.0077	0.3054 ± 0.0225	0.1323 ± 0.0157
Layer 5	$95.33\% \pm 1.15\%$	0.4450 ± 0.0078	0.3080 ± 0.0413	0.1370 ± 0.0354

Table 33: FASHION-MNIST Sandals vs Sneakers – Bottleneck Architecture ($k = 20$)

Layer	**Accuracy**	**Intra-class (κ_{intra})**	**Inter-class (κ_{inter})**	**Geometric Delta ($\Delta\kappa$)**
Layer 1	$88.11\% \pm 2.14\%$	0.4264 ± 0.0183	0.3530 ± 0.0164	0.0734 ± 0.0075
Layer 2	$90.67\% \pm 1.45\%$	0.4336 ± 0.0070	0.3544 ± 0.0098	0.0793 ± 0.0103
Layer 3	$93.89\% \pm 2.59\%$	0.4497 ± 0.0033	0.3407 ± 0.0162	0.1090 ± 0.0133
Layer 4	$94.89\% \pm 2.04\%$	0.4544 ± 0.0072	0.3208 ± 0.0364	0.1336 ± 0.0293
Layer 5	$95.33\% \pm 1.15\%$	0.4585 ± 0.0084	0.3162 ± 0.0365	0.1423 ± 0.0285

Table 34: FASHION-MNIST Sandals vs Sneakers – Collapsed Architecture ($k = 10$)

Layer	**Accuracy**	**Intra-class (κ_{intra})**	**Inter-class (κ_{inter})**	**Geometric Delta ($\Delta\kappa$)**
Layer 1	$88.89\% \pm 0.69\%$	0.3901 ± 0.0090	0.3315 ± 0.0103	0.0586 ± 0.0056
Layer 2	$89.44\% \pm 1.84\%$	0.3926 ± 0.0084	0.3081 ± 0.0077	0.0845 ± 0.0146
Layer 3	$93.56\% \pm 1.68\%$	0.4194 ± 0.0167	0.3069 ± 0.0184	0.1125 ± 0.0246
Layer 4	$94.33\% \pm 0.88\%$	0.4299 ± 0.0172	0.2929 ± 0.0138	0.1370 ± 0.0207
Layer 5	$94.89\% \pm 2.04\%$	0.4137 ± 0.0066	0.3064 ± 0.0136	0.1074 ± 0.0198

Table 35: FASHION-MNIST Sandals vs Sneakers – Collapsed Architecture ($k = 15$)

Layer	**Accuracy**	**Intra-class (κ_{intra})**	**Inter-class (κ_{inter})**	**Geometric Delta ($\Delta\kappa$)**
Layer 1	$88.89\% \pm 0.69\%$	0.3983 ± 0.0153	0.3301 ± 0.0108	0.0682 ± 0.0189
Layer 2	$89.44\% \pm 1.84\%$	0.4010 ± 0.0030	0.3172 ± 0.0124	0.0838 ± 0.0110
Layer 3	$93.56\% \pm 1.68\%$	0.4239 ± 0.0151	0.3165 ± 0.0174	0.1074 ± 0.0247
Layer 4	$94.33\% \pm 0.88\%$	0.4370 ± 0.0177	0.3124 ± 0.0193	0.1246 ± 0.0298
Layer 5	$94.89\% \pm 2.04\%$	0.4271 ± 0.0094	0.3036 ± 0.0214	0.1235 ± 0.0308

Table 36: FASHION-MNIST Sandals vs Sneakers – Collapsed Architecture ($k = 20$)

Layer	**Accuracy**	**Intra-class (κ_{intra})**	**Inter-class (κ_{inter})**	**Geometric Delta ($\Delta\kappa$)**
Layer 1	$88.89\% \pm 0.69\%$	0.4138 ± 0.0119	0.3453 ± 0.0102	0.0685 ± 0.0191
Layer 2	$89.44\% \pm 1.84\%$	0.4167 ± 0.0037	0.3337 ± 0.0131	0.0830 ± 0.0095
Layer 3	$93.56\% \pm 1.68\%$	0.4387 ± 0.0115	0.3275 ± 0.0153	0.1112 ± 0.0232
Layer 4	$94.33\% \pm 0.88\%$	0.4516 ± 0.0175	0.3254 ± 0.0197	0.1261 ± 0.0338
Layer 5	$94.89\% \pm 2.04\%$	0.4437 ± 0.0086	0.3082 ± 0.0168	0.1355 ± 0.0252